Managing the Cholestatic Infant, Whatever the Cause: Ursodeoxycholic Acid, Vitamins, Itch, Growth and Monitoring.

Picture a seven-week-old boy who had a Kasai operation for biliary atresia three weeks ago. His stools have some colour, but his total bilirubin is still a little above three milligrams per deciliter. He is on ursodeoxycholic acid, the bile acid drug most of us simply call urso, and on a combined vitamin drop. His weight has crept up along the fifth centile, and the family is relieved. On examination his liver is firm, his spleen is just palpable, and the skin over his upper arms is loose. Nobody has measured a vitamin level yet. Nobody has measured his arm. And nobody has asked whether the urso he is taking has ever been shown to change anything that matters.

The clinical question of this chapter is what every cholestatic infant needs from us once the diagnosis has been made, or while it is still being made, regardless of the cause. The main idea is that cholestasis is itself a disease of the gut and of nutrition. Whatever has gone wrong in the liver, too little bile acid reaches the intestine, and the consequences follow the same chain in every infant: fat is lost, the fat-soluble vitamins are lost, energy runs short, the bones soften, the nerves suffer, the blood fails to clot, and the child does not grow. The supportive programme that answers this chain decides how well the child arrives at the Kasai, how long the native liver lasts, and how fit the child is if a transplant becomes necessary. The second main idea is less comfortable. Much of what we routinely prescribe, urso above all, rests on habit and on biochemical surrogates rather than on trials with outcomes that families would recognise.

A note on sources. The diagnostic framework is the 2017 joint guideline on cholestatic jaundice in infants written by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. That guideline deals with diagnosis and is almost silent on supportive care. The practical targets come instead from the 2019 joint NASPGHAN and ESPGHAN position paper on nutrition support in chronic liver disease, written by Mouzaki and colleagues. The European Association for the Study of the Liver, known as EASL, published a clinical practice guideline on genetic cholestatic liver diseases in 2024, which rates the drugs for itch and makes the main statements about urso. The vitamin data come from the Childhood Liver Disease Research Network study of Shneider and colleagues, published in 2012. The growth data come from DeRusso and colleagues in 2007 and from Utterson and colleagues in 2005. The itch trials are ICONIC, ASSERT, PEDFIC1 and MARCH, published between 2021 and 2024, and the trials in biliary atresia called EMBARK and BOLD. In 2026 the American Association for the Study of Liver Diseases, the American Society of Transplantation and NASPGHAN published a new guideline on evaluating children for liver transplantation.

Part one. Why cholestasis starves the infant.

Start with the physiology, because everything else follows from it. Fat in milk is mostly long-chain triglyceride. To be absorbed, it has to be broken down by pancreatic lipase and then carried to the intestinal wall inside micelles, which are tiny packages built from bile acids. Micelles form only when the concentration of bile acids in the intestinal lumen is above a threshold called the critical micellar concentration. In a cholestatic infant, whatever the cause, the bile acid concentration in the gut falls below that threshold. Without micelles, long-chain fat is poorly absorbed, and so are the four vitamins that dissolve in fat, which are vitamins A, D, E and K.

The chain then runs in five steps. First, bile flow fails. Second, micelles fail to form. Third, fat and vitamins are malabsorbed, with steatorrhoea and loss of the essential fatty acids that the body cannot make. There is one escape route here. Medium-chain triglyceride, usually shortened to MCT, is absorbed directly into the portal blood without needing micelles, which is why MCT-rich formulas are the backbone of feeding in these infants. Fourth, the infant becomes catabolic. Energy expenditure rises, appetite falls, and an enlarged liver and spleen, sometimes with ascites, press on the stomach and cause early satiety. Fifth come the deficiencies themselves: rickets and fractures from lack of vitamin D, night blindness and poor immunity from lack of vitamin A, a progressive neuropathy from lack of vitamin E, bleeding from lack of vitamin K, and poor growth from all of it together.

The joint nutrition paper translates this into a few numbers worth remembering. The energy target in a cholestatic infant is one hundred and thirty percent of the normal requirement for age. Fat should provide thirty to fifty percent of calories, starting with medium-chain and long-chain fat in a ratio of about thirty to seventy, and medium-chain fat should stay below eighty percent of the total, because too much of it starves the child of essential fatty acids. Linoleic acid should provide at least three percent of calories. And, to show how quickly things go wrong, in the Shneider study fifty-eight percent of infants had at least one vitamin insufficiency only one month after their Kasai, despite taking a supplement designed for exactly this problem.

Part two. Ursodeoxycholic acid: what it can and cannot do.

Urso is a hydrophilic bile acid, meaning that it is more water-friendly and less damaging to cell membranes than the bile acids that accumulate in cholestasis. Given by mouth, it enriches the bile acid pool, it stimulates a flow of bile rich in bicarbonate, and it protects bile duct cells from the detergent injury of the more toxic bile acids. All of these actions depend on one thing: urso must be taken up by the liver and secreted into bile. That tells you where it can and cannot help. It cannot open a bile duct that has been destroyed. It cannot replace a missing transporter, such as the bile salt export pump, abbreviated BSEP, whose absence causes one form of progressive familial intrahepatic cholestasis, abbreviated PFIC. It cannot correct a defect in bile acid synthesis. Where it has a real target is in a disease called multidrug resistance protein three deficiency, caused by variants in the gene ABCB4, in which the liver cannot secrete enough

phospholipid into bile. If some function remains, urso dilutes the toxic bile and helps.

There is a warning to attach before the first dose. The only mention of urso in the 2017 diagnostic guideline is that urinary bile acid screening should be considered before urso is started. The reason is that the inborn errors of bile acid synthesis, which typically cause cholestasis with a normal gamma-glutamyl transferase, the bile duct enzyme usually called GGT, are among the few causes of infant cholestasis that are curable, by giving primary bile acids such as cholic acid. Once urso is circulating, the urine bile acid profile measured by mass spectrometry becomes harder to read. So the order is fixed: send the urine first, then start the drug.

Now the evidence, cause by cause, and it is thinner than our prescribing suggests. In biliary atresia after a Kasai, there is no randomised trial of urso at all. The best data come from a withdrawal study published in 2008 by Willot and colleagues. Sixteen children with a successful Kasai who had taken urso for at least a year, at a mean dose of twenty-five milligrams per kilogram per day, had it stopped. In thirteen of the sixteen, liver tests worsened. The alanine aminotransferase, or ALT, rose from about one point four times the upper limit of normal to three times. The GGT rose from about four point two times to eight times. The aspartate aminotransferase rose more modestly, from one point three to one point seven times. When urso was restarted, all of them improved. In other words, the bar chart of this study shows each enzyme roughly doubling off the drug and falling back on it. This tells us that urso moves enzymes in drained biliary atresia. It does not tell us that it changes how long the child keeps the native liver, and no study has shown that.

In preterm infants with cholestasis associated with parenteral nutrition, a 2024 meta-analysis by Anne and colleagues gathered five studies, three of them randomised. Urso given to prevent cholestasis did not reduce cholestasis, death, sepsis or necrotising enterocolitis. The peak conjugated bilirubin fell by only about one tenth of a milligram per deciliter, and the certainty of the evidence was low to very low. In cystic fibrosis liver disease, the 2017 Cochrane review by Cheng and colleagues found four trials with one hundred and thirty-seven participants and no change in liver enzymes, death or transplantation, and concluded there was insufficient evidence to justify routine use. In ABCB4 disease, by contrast, a French cohort of thirty-eight patients reported by Gonzales and colleagues in 2023 found that about eighty-seven percent of patients with at least one missense variant, meaning a variant that changes one amino acid and may leave some function, responded, whereas none with severe genotypes did, and response predicted longer survival with the native liver. For the other forms of PFIC and for Alagille syndrome there are no controlled data. Finally, a 2023 meta-analysis by Huang and colleagues pooled thirty-two randomised trials in more than two thousand children with cholestasis of many causes. Symptoms improved, with a relative risk of about one point two four, and enzymes and bilirubin fell, but the evidence was short-term, largely about surrogates, and of low certainty. Gastrointestinal side effects occurred in about one child in ten and were no more common than with control treatment.

Put these together and the summary is simple. Outside ABCB4 disease, every positive signal for urso is a biochemical surrogate. EASL states that urso is not generally considered first-line treatment in the genetic cholestases because of lack of evidence, recommends it strongly for ABCB4 disease with at least one missense variant, and considers the data insufficient to advise for or against it in alpha-one antitrypsin deficiency.

There is also a ceiling on the dose. In adults with primary sclerosing cholangitis, a randomised trial published by Lindor and colleagues in 2009 compared a high dose of twenty-eight to thirty milligrams per kilogram per day with placebo in one hundred and fifty patients. The high dose did harm. Thirty-nine percent of patients on urso reached the composite primary endpoint of cirrhosis, varices, cholangiocarcinoma, transplant listing or death, against twenty-six percent on placebo, a risk about two point three times higher. Serious adverse events occurred in sixty-three percent versus thirty-seven percent. The proposed mechanism, which remains unproven, is that excess urso reaching the colon is converted by bacteria into lithocholic acid, a bile acid that is toxic to the liver. These are adult data in a different disease, and nobody has measured this in infants. But they are enough to argue for keeping the dose in the usual range of ten to twenty milligrams per kilogram per day and against escalating the dose because a bilirubin will not fall. Notice that the dose range in the Willot withdrawal study reached thirty-six milligrams per kilogram, which overlaps the range that harmed adults.

Part three. The fat-soluble vitamins.

The most important single study here is the one by Shneider and colleagues from the Childhood Liver Disease Research Network, published in 2012. Ninety-two infants who had had a Kasai were given a liquid multivitamin containing a special water-soluble form of vitamin E called TPGS, which stands for D-alpha-tocopheryl polyethylene glycol succinate, together with extra vitamin K three times a week. TPGS forms its own micelles and was expected to carry the other vitamins in with it. The investigators then measured all four vitamins at one, three and six months, using clear thresholds. Vitamin A was insufficient if the molar ratio of retinol to retinol-binding protein was below zero point eight. Vitamin D was insufficient if the twenty-five hydroxy vitamin D level was below fifteen nanograms per milliliter. Vitamin E was insufficient if the ratio of vitamin E to total serum lipid was below zero point six milligrams per gram, a correction needed because cholestatic infants have high blood lipids that make a raw vitamin E level look falsely normal. Vitamin K was insufficient if the international normalised ratio, the INR, was above one point two.

The result was striking. At least one vitamin was insufficient in fifty-eight percent of infants at one month, fifty-three percent at three months and fifty-seven percent at six months. The decisive variable was bilirubin. At six months, among infants whose total bilirubin was two milligrams per deciliter or more, vitamin A was insufficient in all of them, one hundred percent, compared with nineteen percent of those whose bilirubin was below two. Vitamin D was insufficient in seventy-nine percent versus three percent. Vitamin E was insufficient in fifty

percent versus none. Vitamin K was insufficient in forty-six percent versus six percent. The bar chart of these numbers shows tall red bars for the infants with high bilirubin and almost nothing for those whose jaundice had cleared. Most telling of all, among infants with bilirubin of two or more at six months, not one of twenty-three was sufficient in all four vitamins. The lesson is that a combined product containing TPGS is a reasonable start, but once bilirubin stays above two milligrams per deciliter, each vitamin must be dosed separately and each must be measured.

The joint nutrition paper gives the working doses. For vitamin A, five thousand international units a day in infants under ten kilograms and ten thousand units a day above ten kilograms, with a serum retinol below twenty micrograms per deciliter prompting a test called the modified relative dose response. Retinol-binding protein becomes unreliable in advanced liver disease, and too much vitamin A itself causes liver fibrosis, so this is a vitamin to measure rather than to push blindly. For vitamin D, cholecalciferol two thousand to five thousand units a day, with no agreed upper target. For vitamin E, TPGS fifteen to twenty-five units per kilogram per day, judged by the vitamin E to total lipid ratio, with deficiency defined as below zero point six milligrams per gram from one to twelve years of age and below zero point eight in older children. For vitamin K, two to five milligrams a day by mouth, with one to ten milligrams intravenously when needed, remembering that intravenous vitamin K carries a small risk of anaphylaxis. In the Shneider study, doses were escalated well beyond these starting points when levels stayed low: vitamin A up to twenty-five to fifty thousand units a day, or fifty thousand units by intramuscular injection monthly; vitamin D up to eight thousand units a day, or the active form calcitriol; TPGS up to one hundred units per kilogram per day; and vitamin K by INR, two and a half milligrams a day for an INR between one point two and one point five, and an intramuscular dose plus daily oral vitamin K above that. Levels should be checked every three to six months, and monthly in a severely cholestatic infant. Essential fatty acids should be checked every three to six months using a red-cell fatty acid profile, because the older triene to tetraene ratio alone is not adequate.

Vitamin K deserves a section of its own, because its deficiency can be the first sign of the disease and it can be catastrophic. Across three British Paediatric Surveillance Unit surveys reported by McNinch and colleagues, nineteen cases of late vitamin K deficiency bleeding occurred in infants with underlying liver disease, six in the first survey, twelve in the second and one in the third. Most were breast-fed infants who had received no prophylaxis or a single oral dose at birth. Breast milk contains little vitamin K, and an infant who cannot absorb fat cannot absorb the oral vitamin either. A Japanese study of two hundred and eighty-nine infants with biliary atresia born between 2015 and 2019, published in 2025, found no vitamin K deficiency bleeding among those given a three-month oral prophylaxis regimen, and more bleeding with a shorter three-dose regimen. In practice, check the INR on the first day in every jaundiced infant and give parenteral vitamin K if it is prolonged. If the INR corrects within hours, the problem was deficiency rather than failure of the liver to make clotting factors. Afterwards, give daily oral vitamin K and recheck the INR at every visit. And remember that a normal INR does not prove adequate vitamin K stores for bone; a test called PIVKA-two, which measures

under-carboxylated prothrombin, detects deficiency earlier where it is available. The trap to avoid is the breast-fed infant whose single oral dose at birth was taken as protection, and whose first sign of biliary atresia or alpha-one antitrypsin deficiency is a bleed inside the head.

Part four. Nutrition and growth.

The joint paper sets the targets. Energy should be one hundred and thirty percent of the age requirement, and at least that in end-stage disease, measured by indirect calorimetry where available. Protein should be one hundred and thirty to one hundred and fifty percent of the age requirement. Protein should not be restricted, and any restriction for severe encephalopathy should last no longer than two to three days. Note that the paper expresses protein as a share of the age requirement, not as grams per kilogram. Fat should provide thirty to fifty percent of calories, with medium-chain fat starting at about thirty percent of fat and staying below eighty percent; tube formulas with about half their fat as medium-chain fat may be used. Linoleic acid should provide at least three percent of calories, and alpha-linolenic acid zero point seven to one percent. A single paediatric randomised trial found that a formula enriched with branched-chain amino acids, given by nasogastric tube, improved body measurements, but no dose is recommended. Tube feeding should begin if energy goals are not met for more than two weeks, or if growth is poor for two to four weeks. Infants can be fed every one to two hours or continuously, and a percutaneous gastrostomy should be avoided when there is portal hypertension, because of varices in the stomach wall. In practice, breast milk alone rarely meets these targets in an infant with significant cholestasis, and fortification with an MCT-rich formula is usually needed.

Why does this matter so much? Because growth predicts outcome. DeRusso and colleagues followed one hundred infants with biliary atresia from nine centres in the United States after their Kasai. Forty-six had a poor outcome, meaning transplantation or death by twenty-four months. Six months after the Kasai, their mean weight-for-age z-score, which counts how many standard deviations a child sits below the average, was minus two point one, compared with minus one point two in the children who did well, a highly significant difference. Even among infants whose bilirubin was in the same middle range of two to six milligrams per deciliter at three months, poor growth still separated those who did badly, with z-scores of minus two versus minus one. In the transplant registry analysed by Utterson and colleagues, which included seven hundred and fifty-five listed children with biliary atresia, a height or weight more than two standard deviations below the mean predicted death after transplant and loss of the graft. The practical conclusion is that growth faltering after a Kasai is both a liver sign and a nutritional emergency. Start tube feeding and refer for transplant assessment in parallel, not in sequence.

There is a catch in how we measure growth. Weight lies in these children. An enlarged liver and spleen, and any ascites, add weight that is not tissue, so weight can rise while muscle and fat are melting away. The joint paper says that length is more meaningful than weight, and recommends

serial measurement of the mid-upper arm circumference and the triceps skinfold, every two to four weeks in cholestatic infants. The arm is free of organomegaly, so it shows wasting early. The triceps skinfold measures fat stores directly, and together the two allow an estimate of arm muscle. Head circumference belongs on every visit as well, and development is an outcome in its own right, partly because vitamin E neuropathy is silent until it is advanced. The pitfall to remember is the infant whose weight z-score is improving while the spleen grows and ascites appears. That is not catch-up growth. Look at the arm before you relax.

Part five. Itch.

Cholestatic itch, or pruritus, is not a cosmetic symptom. Infants scratch until they bleed, they and their parents lose sleep, and in some genetic cholestases intractable itch is itself the reason for transplantation. Its cause is not fully understood, but retained bile acids and other substances in the circulation play a central part, which is why interrupting the circulation of bile acids helps.

A practical ladder runs as follows. Begin with skin care and sleep: emollients, short nails, a cool room, and a sedating antihistamine at night only to help sleep, since antihistamines do little for cholestatic itch itself and EASL rates their efficacy as marginal. The next step is rifampicin, an antibiotic that also induces liver enzymes which detoxify itch-causing substances. In the 1993 series of Gregorio and colleagues in thirty-three children, at a median dose of five milligrams per kilogram per day, itch resolved completely in fifteen percent and partly in thirty-six percent. EASL rates its efficacy as moderate, and liver tests should be checked because rifampicin can itself injure the liver. The next step is sertraline, an antidepressant. In a 2017 French series by Thébaut and colleagues of twenty children with Alagille syndrome or PFIC whose itch persisted despite urso and rifampicin, sertraline at one to four milligrams per kilogram per day, with a median of two point two, improved itch in fourteen of twenty, with itch scores falling on average from eight out of ten to five out of ten. Six children had side effects, including agitation, skin reactions, hair loss and vomiting. Cholestyramine, a resin that binds bile acids in the gut, and naltrexone, which blocks opioid receptors, are rated as marginal by EASL and are awkward in infants; cholestyramine also binds the fat-soluble vitamins and urso, so the doses have to be separated by several hours.

The newest step is a class of drugs called ileal bile acid transporter inhibitors, usually abbreviated IBAT inhibitors. In the last part of the small intestine, the ileum, a transporter recovers about ninety-five percent of bile acids and returns them to the liver. Block it, and more bile acids are lost in the stool, the circulating pool shrinks, and itch often improves. Two drugs are in use, maralixibat and odevixibat, and four randomised trials define them. In the ICONIC trial in Alagille syndrome, published by Gonzales and colleagues in 2021, thirty-one children took maralixibat and were then randomly withdrawn to placebo or continued; on placebo, serum bile acids rose by about ninety-four micromoles per liter and itch scores rose by one point seven points, and benefit was maintained for up to two hundred and four weeks in those who continued. In the ASSERT trial in Alagille syndrome, fifty-two children were randomised to

odevixibat or placebo for twenty-four weeks; the scratch score fell by one point seven points against zero point eight on placebo, and bile acids fell by ninety micromoles per liter while they rose by twenty-two on placebo. In the PEDFIC1 trial in PFIC types one and two, sixty-two children took odevixibat or placebo for twenty-four weeks. Positive itch assessments occurred fifty-five percent of the time against thirty percent on placebo, and a bile acid response occurred in thirty-three percent against none. Diarrhoea occurred in thirty-one percent against ten percent, while serious adverse events were actually less frequent on the drug, seven percent against twenty-five percent. The bar chart of PEDFIC1 therefore shows a clear benefit in itch and bile acids, and a clear cost in diarrhoea. In the MARCH trial of maralixibat in ninety-three children with PFIC, the itch score in the BSEP-deficient group fell by one point seven against zero point six on placebo, and diarrhoea occurred in fifty-seven percent against twenty percent. Because diarrhoea means more fat loss, fat-soluble vitamins need closer watching once these drugs start.

Two points limit their use. First, they need bile acids to be secreted in the first place, so they work poorly when transport is absent altogether, and severe BSEP deficiency with no functional protein is excluded from the maralixibat label. Second, they have failed in biliary atresia. The EMBARK trial of maralixibat after a Kasai missed its bilirubin endpoint in 2023, and the BOLD trial of odevixibat in two hundred and fifty-four infants after a Kasai missed its primary endpoint of survival with the native liver, according to the company's report in July 2026. As of 2026, the American labels allow odevixibat for PFIC from three months and for Alagille syndrome from twelve months, and maralixibat for Alagille syndrome from three months and for PFIC from twelve months; the European approvals for odevixibat cover PFIC and Alagille syndrome from six months of age. EASL recommends strongly that an IBAT inhibitor be offered to children with Alagille syndrome and cholestatic itch where available.

At the top of the ladder sit surgical biliary diversion and transplantation. EASL advises against routine diversion in FIC1 deficiency, the form of PFIC caused by ATP8B1 variants, and says it should be considered in BSEP deficiency with a responsive genotype. In the data behind that recommendation, itch responded to diversion in fifty-four percent overall, in sixty-six percent of children carrying the two milder variants called E297G and D482G, in thirty-six percent with other missense variants, and in none of three children with two truncating variants. Measure itch at every visit, with a caregiver scratch score or the observer version of the itch-reported outcome instrument, because sleep is the outcome families feel most.

Part six. Surveillance, and what the guidelines say.

A practical monitoring schedule follows from everything above. Growth, including weight, length, head circumference, arm circumference and triceps skinfold, every two to four weeks in infancy. Vitamin levels every three to six months, monthly in the severely cholestatic infant. Essential fatty acids every three to six months, and urea and creatinine every three months. At each visit, the size of the spleen and the platelet count, because a falling platelet count with a growing spleen is the earliest sign of portal hypertension, with Doppler ultrasound as the course

requires. Pulse oximetry, upright and lying down, to screen for hepatopulmonary syndrome, in which dilated blood vessels in the lungs let blood bypass oxygenation. In one series of one hundred and twenty children with biliary atresia, hepatopulmonary syndrome was present in seven percent, and eight of the ten cases had biliary atresia; it is silent until an oximeter finds it, and a saturation below ninety-seven percent should prompt investigation. And readiness for transplant: the 2026 North American transplant evaluation guideline recommends aggressive nutritional support before transplant, an accelerated vaccine schedule with measles, mumps, rubella and varicella vaccines from six months of age, because live vaccines cannot be given after transplant, screening for hepatopulmonary syndrome by oximetry, and allocation using the paediatric end-stage liver disease score that now includes creatinine, called PELD-Cr, adopted in 2023.

When we compare the documents, the picture is mixed. On urso for cholestatic infants in general, the North American and European positions are aligned, in the sense that both are either silent or sceptical: Fawaz makes no recommendation beyond screening for synthesis defects first, and EASL says urso is not first line. On urso after a Kasai, there is a gap: no guideline grades it, although its use is nearly universal. On urso in ABCB4 disease, the documents differ, because only EASL recommends it. On vitamin doses and markers, and on energy, protein and medium-chain fat, the documents are aligned, because both regions signed the same 2019 joint paper. On IBAT inhibitors they differ, mainly in licensed ages. On biliary diversion there is a North American gap. And on transplant evaluation there is a material difference, since the 2026 North American guideline has no European paediatric equivalent. The single most material gap is that no guideline grades the use of urso after a Kasai, which is the most common prescription in this population.

Pearls and pitfalls.

Four traps recur. The first is to believe that the combined multivitamin covers it; it fails once bilirubin is above two. The second is to start urso first and test later, which can hide a curable synthesis defect. The third is to be reassured that the infant is gaining weight, when the weight is liver, spleen and fluid. The fourth is to assume that a normal INR means enough vitamin K, when the INR is a late marker. Two more mistakes deserve mention: escalating urso above twenty milligrams per kilogram for a bilirubin that will not fall, and starting an IBAT inhibitor after a Kasai outside a trial. Neither has evidence of benefit, and the first carries an adult signal of harm.

Let us return to the boy from the beginning, and then to a few others. For the seven-week-old after his Kasai, with a bilirubin above three on a combined drop, measure vitamins A, D and E and the INR now, add separate vitamin D, TPGS and daily vitamin K, recheck in four to six weeks, measure his arm and his length, and move him onto an MCT-rich feed with a low threshold for a nasogastric tube. For a six-week-old whose cause is still unknown and whose team wants to start urso, send urinary bile acids by mass spectrometry first and start urso at ten

to twenty milligrams per kilogram afterwards if it is still indicated. For a breast-fed seven-week-old who had a single oral dose of vitamin K at birth and now is irritable with an INR of six, treat as vitamin K deficiency bleeding, with intravenous vitamin K and urgent head imaging, and start the cholestasis work-up. For an infant after a Kasai who is gaining weight while his spleen grows and his arm circumference falls, recognise wasting hidden under organomegaly, start nasogastric MCT-rich feeds and refer for transplant assessment. For a fourteen-month-old with Alagille syndrome who scratches through the night despite rifampicin, offer an IBAT inhibitor and watch for diarrhoea and falling vitamin levels. For a child with PFIC type two and two truncating ABCB11 variants with refractory itch, expect neither an IBAT inhibitor nor diversion to work, and plan for transplant. And for a nine-month-old with biliary atresia whose oxygen saturation is ninety-five percent lying down, investigate for hepatopulmonary syndrome, because it changes transplant timing.

What to do differently in clinic.

So, in clinic tomorrow, treat the cholestasis and not only its cause. Send urinary bile acids before the first dose of urso, keep urso at ten to twenty milligrams per kilogram per day, and be honest with families that it improves blood tests rather than proven outcomes, except in ABCB4 disease. Once bilirubin stays above two, stop relying on a combined vitamin drop: dose vitamins A, D, E and K separately, measure each of them every three to six months, and give daily vitamin K with an INR at every visit. Feed to one hundred and thirty percent of energy needs with medium-chain fat, and reach for a nasogastric tube after two weeks of missed goals rather than two months. Measure the arm and the length, not just the weight, and treat a falling growth curve as a reason to call the transplant team. Climb the itch ladder deliberately, from rifampicin to sertraline to an IBAT inhibitor where the diagnosis supports it, and do not offer those drugs after a Kasai outside a trial. And check an oxygen saturation, upright and lying down, at every visit.